

The experimental objective in May:

Module 1: Construct of BW-pYB1a-*vdh-nox* and test HPLC methods、protein expression

Module 2: Construct an *odm* mutant library with qualified mutation rate.

Module 3: Construct BW-pLB1s-*anAB-aroY*, BW-pYB1a-*dmpB-dmpD-mhpD-mhpE* and test HPLC methods

Week 1 (5.15-5.20)

Module 1:

1.Inoculation

Original strains: BW25113、BW-pYB1a-*eGFP*、BW-pYB1a-*vdh*、BW-pYB1a-*progo*、BW-pYB1a-*novIf*

Cultivation in a shaker at 37°C for 12 hours

2.Induction

Induction of BW-pYB1a-*vdh*、BW25113、BW-pYB1a-*progo*、BW-pYB1a-*novIf*、BW-pYB1a-*eGFP*

Induction System	Volume
ZY	4.8 mL
50x 5052 supplement	100 μL
50x M supplement	100 μL
MgSO4	10 μL
1000x trace element	10 μL
Bacterial suspension	50 μL
20% arabinose	50 μL
Antibiotic	5 μL

Cultivation in a shaker at 30°C for 16 hours.

3.Whole-Cell Catalysis

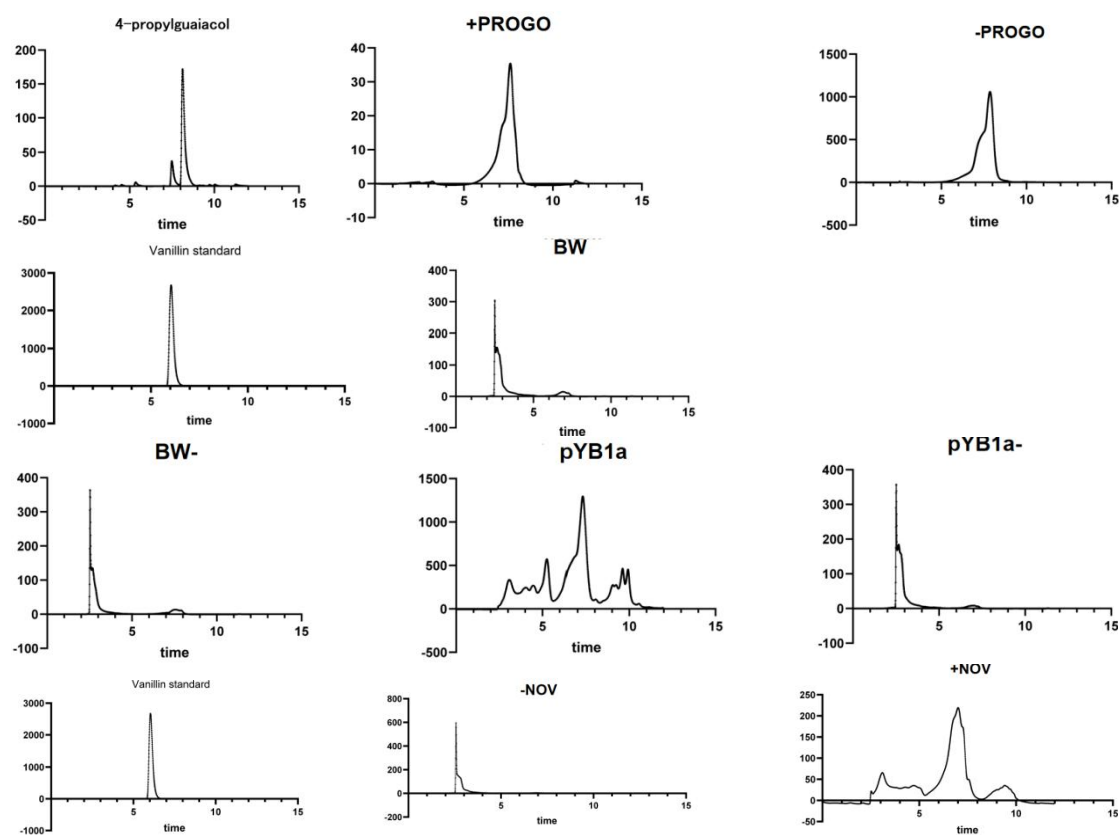
BW-pYB1a-*vdh* catalyzes vanillin-to-VA conversion via induced enzymes. BW-pYB1a-progo catalyzes 4-PG-to-isoeugenol conversion via induced enzymes. BW-pYB1a-nov1f catalyzes isoeugenol-to-vanillin conversion via induced enzymes. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀=6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in pH=7.5 Tris-HCl buffer, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 1 hours.

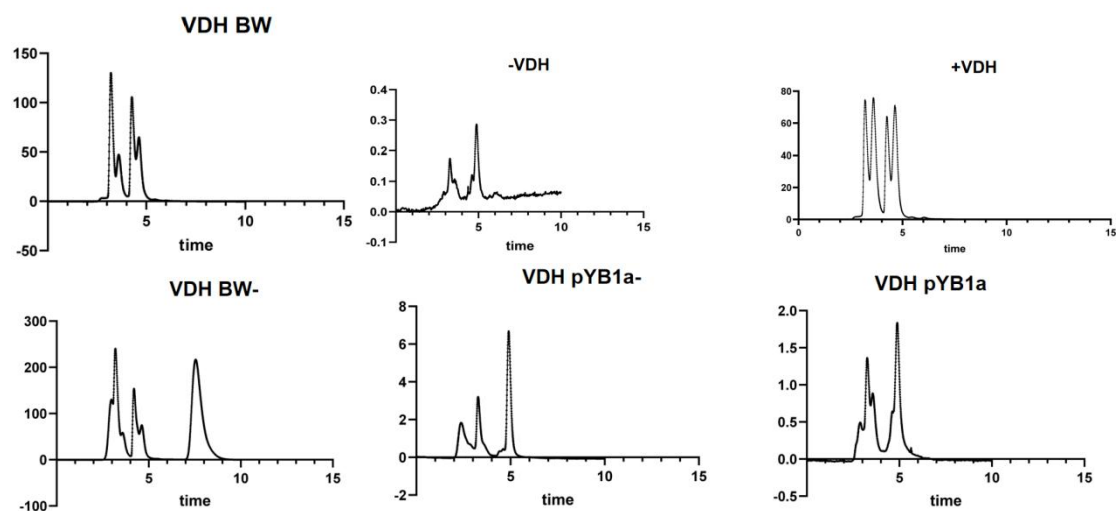
4.High-Performance Liquid Chromatography (HPLC) Detection of VA 、 isoeugenol、 vanillin

The mobile phase was prepared: aqueous phosphoric acid and acetonitrile.

The standard solutions were prepared, and the samples were filtered through 0.22μm membranes for HPLC.

The samples were detected by HPLC using the aqueous phosphoric acid and acetonitrile method. Column: Aminex XP-C18; flow rate: 0.7 mL/min; injection volume: 10μL





Module 2:

(A) Error-prone PCR amplification of *odm*

First- round error-prone PCR amplification of *odm*

1.Error-prone PCR

Error-prone PCR was performed with two different Mg^{2+}/Mn^{2+} ratios to introduce random mutations into ODM coding sequence.

Mg ²⁺ : Mn ²⁺ =10:1		Mg ²⁺ : Mn ²⁺ =5:1	
PCR system (50μL)		PCR system (50μL)	
Template	5ng	Ttemplate	5ng
F	2ul	F	2ul
R	2ul	R	2ul
10 mM Mg ²⁺ solution	1.5μL	10 mM Mg ²⁺ solution	1.5μL
1 mM Mn ²⁺ solution	1.5μL	1 mM Mn ²⁺ solution	3μL
Green Mix	25μL	Green Mix	25μL
DDW	11.8μL	DDW	10.3μL

2.Connection

Connecting the two fragments using the Golden- Gate method.

PCR Program					
98°C		5min			
98°C		30s			
57°C		30s			
72°C		30s			
72°C		5min			
25°C		∞			

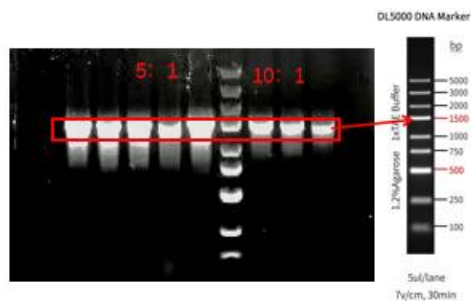
Goden Gate system (10μL)	10:1	5:1	Connection		
			temperature	time	cycles
template	200ng	100ng	37°C	30min	
10×T4 buffer	1 μL	1 μL	37°C	3min	25 X
BsaI	0.5 μL	0.5 μL	16°C	3 min	25 X
T4 DNA ligase	0.3 μL	0.3 μL	16°C	20min	
rCutSmart (10×)	1μL	1μL	37°C	20min	
vector	1 μL	1 μL	55°C	15min	
DDW	10 μL	10 μL	80°C	15min	
			25°C	∞	

3. Transformation

The ligated system was transferred into DH5 α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;

4. Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 1508bp



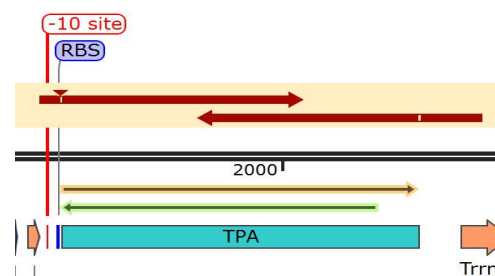
5. Plasmid extraction + send for sequencing

The correctly digested protoplasmid was sent for testing.

10:1

5:1 No mutations detected

```
T A C C A T T A A T T G G G A   T G T T
T A C A T T T A A T T G G G A A T G T T
T A C C A T T A A T T G G G A   T G T T
```



Second- round error- prone PCR amplification of *odm*

1. Error- prone PCR

Error-prone PCR was performed with two different Mg²⁺/Mn²⁺ ratios to introduce random mutations into ODM coding sequence.

Mg²⁺ : Mn²⁺ =10:1

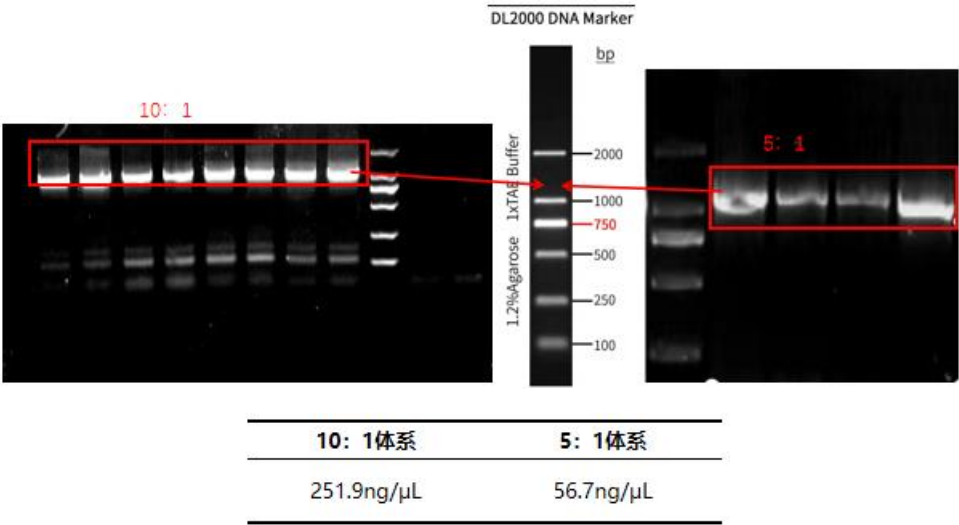
PCR system (50μL)	
template	2.5ng
F	2ul
R	2ul
10 mM Mg ²⁺ solution	1.5μL
1 mM Mn ²⁺ solution	1.5μL
Green Mix	25μL
DDW	14.9μL

Mg²⁺ : Mn²⁺ =5:1

PCR system (50μL)	
template	2.5ng
F	2ul
R	2ul
10 mM Mg ²⁺ solution	1.5μL
1 mM Mn ²⁺ solution	3μL
Green Mix	25μL
DDW	13.4μL

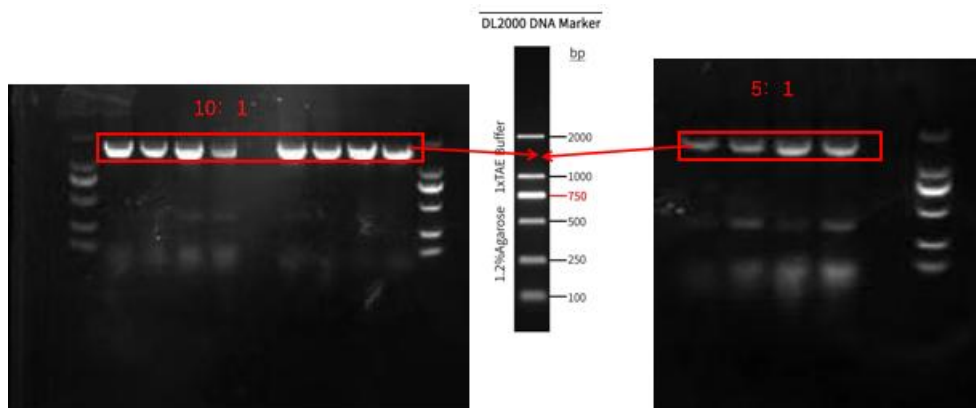
2.Conducted gel extraction for the fragments.

Expected band size: 1133 bp



3.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 1508bp.



(B) Characterization assay of the PobR biosensor

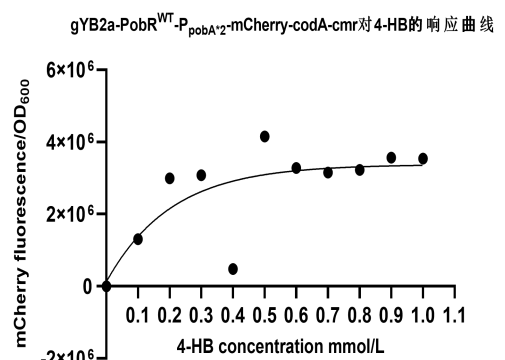
First round of characterization assay of the PobR biosensor

Basal setup (no 4- HB): 1 μ L bacterial culture, 0.2 μ L Amp stock. 200 μ L total of 4- HB gradient in 1 $\times$ M9. Tested concentrations: 0–1.0 mmol/L.

96- well lplate setup:

response curve:

4-HB浓度 mmol/L	1mmol/L 4- HB (ul)	1 $\times$ M9 (ul)
0	0	200
0.1	20	180
0.2	40	160
0.3	60	140
0.4	80	120
0.5	100	100
0.6	120	80
0.7	140	60
0.8	160	40
0.9	180	20
1.0	200	0



Module 3:

1.Design primers

According to the target gene sequences required for subsequent PCR amplification (e.g.*VanAB*, *aroY*), specific amplification primers were designed and synthesized.

2.Inoculation and shake culture

50 μ L of the bacterial preservation solution was transferred to 5mL of liquid LB medium and cultivated in a shaking incubator at 37°C for 12 hours.

3.Plasmid extraction (BW-pLB1s-*aroY*,BW-pYB1a-*VanAB*)

After cultivation, plasmids were extracted from the strains (BW-PLB1S-*aroY*,

BW-PYB1A-*VanAB*), and the plasmid concentrations were measured for subsequent digestion and PCR verification.

4. Enzyme digestion

The extracted plasmids were digested for verification

5. PCR+DNA gel extraction

PCR amplification of the target fragment.

PCR system (50μL):

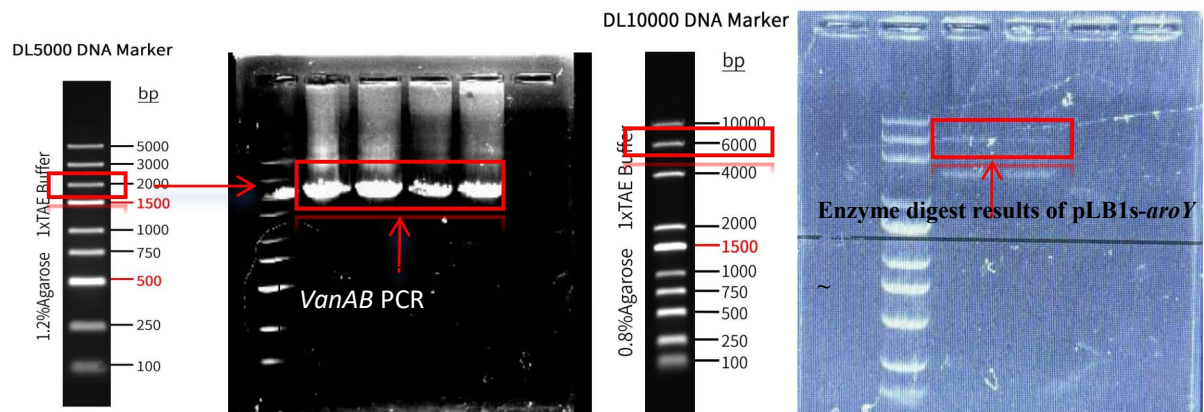
2×Mix	25μL
Primer-F	2μL
Primer-R	2μL
Template	50ng
DDW	20μL

VanAB PCR program:

98°C	5min	} X30
98°C	30s	
58°C	30s	
72°C	65s	
72°C	5min	
25°C	∞	

aroY PCR program:

98°C	5min	} X11
98°C	30s	
72.1°C(-1.1°C)	30s	
72°C	3m30s	
98°C	30s	} X20
60°C	30s	
72°C	3m30s	
25°C	∞	



DNA gel extraction of the target fragment.

After agarose gel electrophoresis, the target fragment band was excised and recovered.

6. Gibson assembly ligation.

Gibson system (10 μ L):

2 $\times$ CEMix	5 μ L
Fragment	1 μ L (diluted)
Vector	1 μ L (diluted)
DDW	to 10 μ L

Gibson: 50 $^{\circ}$ C 15min; 4 $^{\circ}$ C ∞

7. Transformation and plating.

The ligation product was transformed into DH5 α competent cells, plated, and cultured inverted at 37 $^{\circ}$ C for 12 hours.

Due to no colony growth on the plates, repeated the Enzyme digestion, gibson assembly, transformation, and plating steps next week.

Week 2 (5.21-5.27)

Module 1:

1. PCR + DNA gel extraction

The primer was designed, followed by PCR amplification of the target gene and vector pYB1a-*progo*. Subsequently, the relevant DNA fragments were assembled using Gibson Assembly methodology, followed by transformation into competent cells. Upon completion of

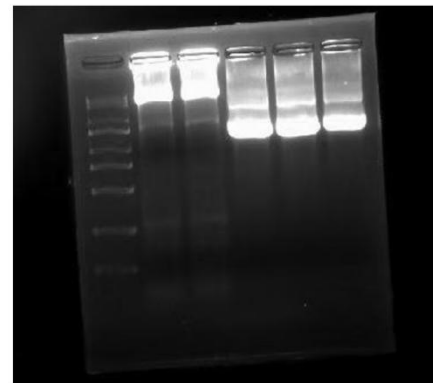
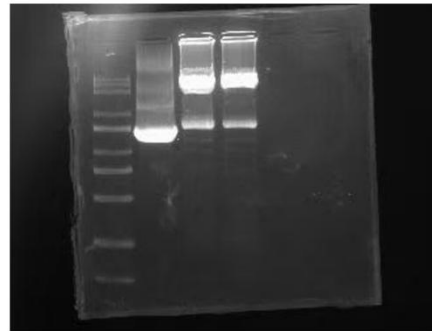
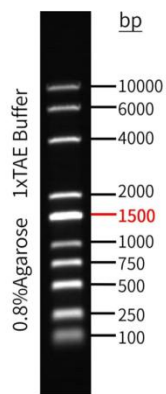
the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed

PCR system (50μL)		PCR system (50μL)	
2×Mix	25ul	2×Mix	25ul
PROGO-F	2ul	NOV1F-F	2ul
PROGO-R	2ul	NOV1F-R	2ul
Template	20ng	Template	20ng
DDW	20ul	DDW	20uL

PCR-pYB1a- <i>vdh</i>		
98°C	5min	} ×30
98°C	30s	
64°C	30s	
72°C	2min33s	
72°C	5min	
25°C	∞	

PCR-pYB1a- <i>nox</i>		
98°C	5min	} ×30
98°C	30s	
64°C	30s	
72°C	45s	
72°C	5min	
25°C	∞	

DL10000 DNA Marker



5ul/lane
7v/cm, 30min

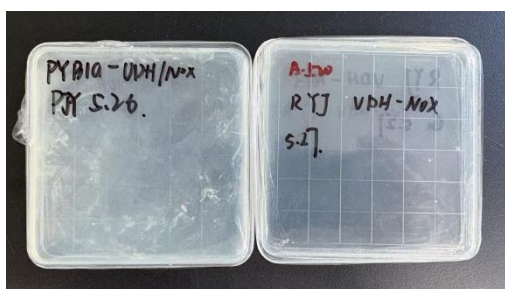
2.Connection

Connecting the two fragments using the Gibson method.

Gibson system (10μL)	
C117	5μL
pYB1a- <i>vdh</i>	98.2g
<i>nox</i>	56.6ng
DDW	1μ
Gibson	
50°C	30min
4°C	∞

3.Transformation

The ligated system was transferred into DH5α and BW25113 receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



4.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 3216 bp

PCR system (10μL)	
2×Mix	5ul
PBF	0.4ul
PBR	0.4ul
Template	1ul
DDW	to 10ul
PCR Program	
98°C	5min
98°C	30s
60°C	30s
72°C	42s
72°C	5min
25°C	∞

} X30



The result showed that colony PCR is failure.

Module 2:

(A) Error-prone PCR amplification of *odm*

1.Third- round error-prone PCR amplification of *odm*

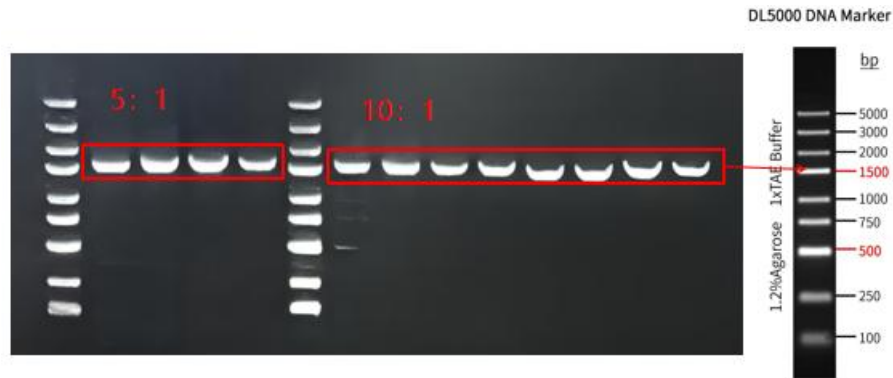
Error-prone PCR was performed with two different Mg^{2+}/Mn^{2+} ratios to introduce random mutations into ODM coding sequence.

$Mg^{2+} : Mn^{2+} = 10:1$		$Mg^{2+} : Mn^{2+} = 5:1$	
PCR system (50 μ L)		PCR system (50 μ L)	
template	1ng	template	1ng
F	2ul	F	2ul
R	2ul	R	2ul
10 mM Mg^{2+} solution	1.5 μ L	10 mM Mg^{2+} solution	1.5 μ L
1 mM Mn^{2+} solution	1.5 μ L	1 mM Mn^{2+} solution	3 μ L
Green Mix	25 μ L	Green Mix	25 μ L
DDW	17 μ L	DDW	15.5 μ L

PCR	
98 $^{\circ}$ C	5in
98 $^{\circ}$ C	30s
56 $^{\circ}$ C	30s
72 $^{\circ}$ C	35s
72 $^{\circ}$ C	5min
25 $^{\circ}$ C	∞

4.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 1508bp.Colony PCR yielded correct target bands, indicating successful ligation.



5. Plasmid extraction + send for sequencing

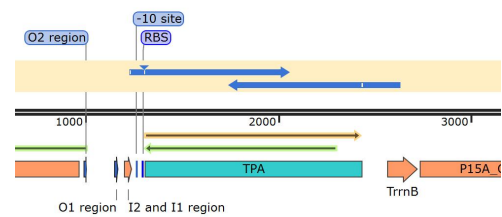
The correctly digested protoplasmid was sent for testing.

10:1

5:1 No mutations detected

```

ATTAAAACCATGCTGAATGGTTT
ATTAAACCCATGCTGAATGGTTT
ATTAAAACCATGCTGAATGGTTT
  
```



Ligation was successful; however, the mutation rate failed to meet requirements. No mutations were obtained after three rounds of error-prone PCR. We suspect that this problem stems from inappropriate ionic concentrations within the error-prone PCR system..

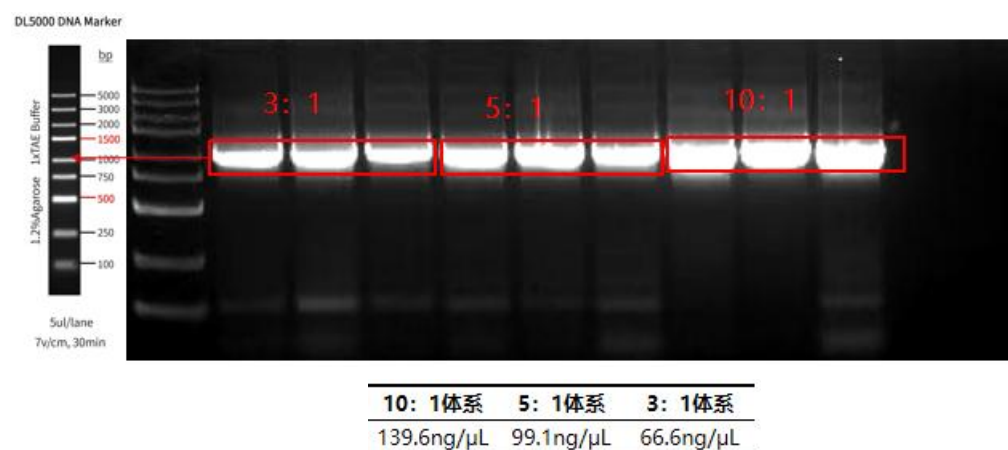
4. Fourth- round error- prone PCR amplification of *odm*

Mg ²⁺ : Mn ²⁺ =10:1		Mg ²⁺ : Mn ²⁺ =5:1	
PCR system (50µL)		PCR system (50µL)	
template	1ng	template	1ng
F	2ul	F	2ul
R	2ul	R	2ul
10 mM Mg ²⁺ solution	1.5 µ L	10 mM Mg ²⁺ solution	1.5 µ L
1 mM Mn ²⁺ solution	1.5 µ L	1 mM Mn ²⁺ solution	3 µ L
Green Mix	25 µ L	Green Mix	25 µ L
DDW	17 µ L	DDW	15.5 µ L

$Mg^{2+} : Mn^{2+} = 3:1$			
PCR system (50 μ L)		PCR	
template	1ng	98°C	5min
F	2ul	98°C	30s
R	2ul	56°C	30s
		72°C	45s
10 mM Mg^{2+} solution	1.94 μ L	72°C	5min
1 mM Mn^{2+} solution	6.45 μ L	25°C	∞
Green Mix	25 μ L	} $\times 35$	
DDW	11.61 μ L		

5. Conducted gel extraction for the fragments.

Expected band size: 1133 bp. The gel- extracted fragment exhibited the correct target band.



6. Connection

Connecting the two fragments using the Golden- Gate method

Connection Program		
temperature	time	cycles
37°C	30min	
37°C	3min	25 X
16°C	3min	25 X

16°C	20min
37°C	20min
55°C	15min
80°C	15min
25°C	∞

Goden-gate system (10μL)	10: 1	5:1
template	10ng	10ng
10×T4 buffer	1 μL	1 μL
BsaI	0.5 μL	0.5 μL
T4 DNA ligase	0.3 μL	0.3 μL
rCutSmart (10×)	1μL	1μL
vector	10ng	10ng
DDW	5.2μL	4.2 μL

(B) Characterization assay of the PobR biosensor

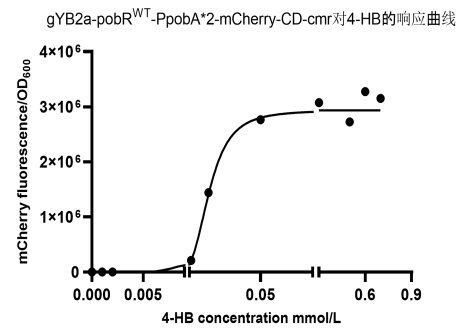
Second- round characterization assay of the PobR biosensor

1.Characterization assay

Basal setup (no 4- HB): 1 μL bacterial culture, 0.2 μL Amp stock. 200 μL total of 4- HB gradient in 1×M9. Tested concentrations: 0–1.0 mmol/L.

96- well plate setup:

4-HB浓度 mmol/L	1mmol/L 4-HB (ul)	1×M9 (ul)
0	0	200
0.001	20	180
	(0.01mmol/L)	
0.002	40	160
	(0.01mmol/L)	
0.008	160	40
	(0.01mmol/L)	
0.01	2	198
0.02	4	196
0.05	10	190
0.1	20	180
0.5	100	100

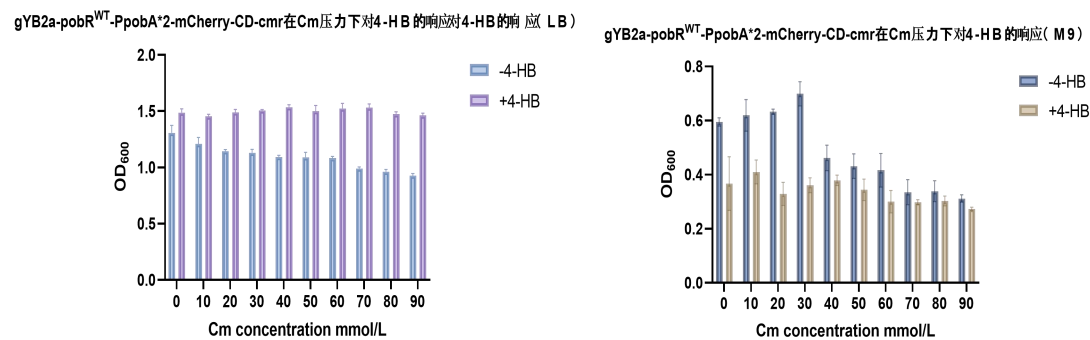


2.Chloramphenicol selection

Basal components (no 4- HB): 1 μ L bacterial culture; 0.2 μ L Amp stock.Each well contained a total volume of 200 μ L with 4- HB, 1 $\times$ M9 medium and chloramphenicol at varying concentrations. A control without chloramphenicol was additionally set up.The optimal stable concentration of 4- HB was fixed at 0.6 mmol/L. The 4- HB stock solution was diluted from 100 mmol/L to 10 mmol/L.Chloramphenicol stock solution was prepared at 0.3 mg/mL. The chloramphenicol concentration gradient was 0, 10, 20, 30, 40, 50, 60, 70, 80, 90 mg/L.

Experimental group / Loading volume (μL) / Chloramphenicol concentration gradient	LB group						1×M9 group				
	4-HB			non-4-HB			4-HB		non-4-HB		
	chlora mphe nicol	4-HB	LB	chlora mphe nicol	LB	chlora mphe nicol	4-HB	1×M9	chlora mphe nicol	1×M9	
1	0	0	12	188	0	200	0	12	188	0	200
2	10	2	12	186	2	198	2	12	186	2	198
3	20	4	12	184	4	196	4	12	184	4	196
4	30	6	12	182	6	194	6	12	182	6	194
5	40	8	12	180	8	192	8	12	180	8	192
6	50	10	12	178	10	190	10	12	178	10	190
7	60	12	12	176	12	188	12	12	176	12	188
8	70	14	12	174	14	186	14	12	174	14	186
9	80	16	12	172	16	184	16	12	172	16	184
10	90	18	12	170	18	182	18	12	170	18	182

In the LB groups, slight differences were observed between samples with and without 4- HB, yet the discrepancy was minor. Bacterial growth showed nearly no variation across different chloramphenicol concentrations.In the M9 groups, bacteria supplemented with 4- HB grew more poorly than those without 4- HB. We suspect this phenomenon arises from leaky expression of the biosensor.



5. Plasmid extraction + send for sequencing

The correctly digested protoplasmid was sent for testing.

10:1

5:1No mutations detected

GCAGCGAACC	GCTGAATGCCAGTGCATTTACCA	GTGTTTAAACCA	CCTGAATGATCTGCTGAACTGCTGGTGGAAATCTGATTAATTGTAAAGAGGTAAACAGCGAATACCATGAAAA
GCAGCGAACC	GCTGAATGCCAGTGCATTTACCA	GTGTTTAAACCA	CCTGAATGATCTGCTGAACTGCTGGTGGAAATCTGATTAATTGTAAAGAGGTAAACAGCGAATACCATGAAAA
GCAGCGAACC	GCTGAATGCCAGTGCATTTACCA	GTGTTTAAACCA	AATGTTGCTGAACTGCTGGTGGAAATCTGATTAATTGTAAAGAGGTAAACAGCGAATACCATGAAAA

3: 1

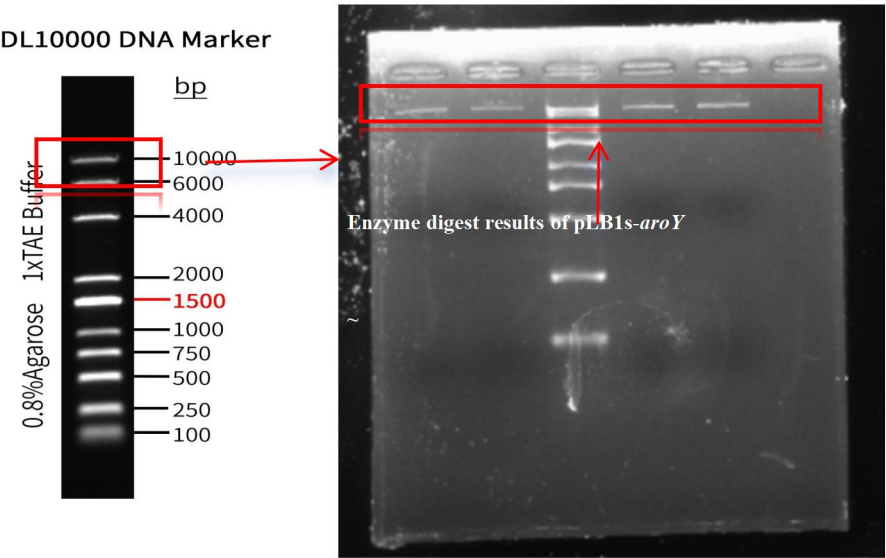
GGAA	CCGATTCTGAACTGACCACCGATTATGATAGC	GTGAGCACCTATTTTAT
GGAA	CCGATTCTGAACTGACCACCGATTATGATAGC	GTGAGCACCTATTTTAT
GGAA	CCGATTCTGAACTGACCACCGATTATGATAGC	GTGAGCACCTATTTTAT

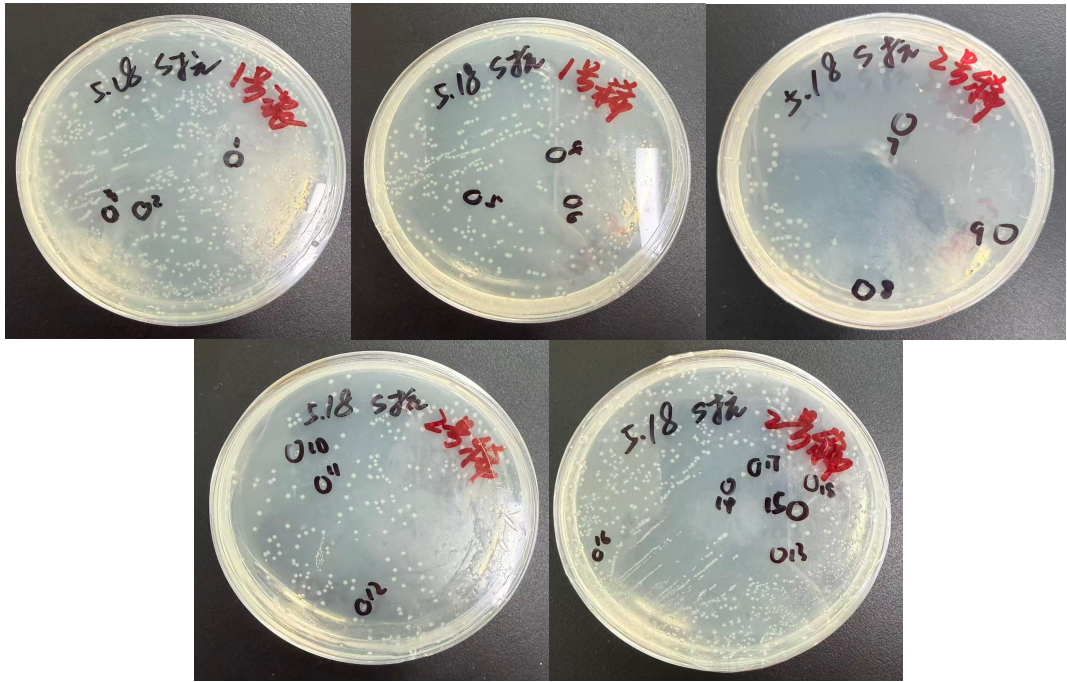
A 7-bp deletion was detected, which was attributed to poor primer quality. Solution: Resynthesize the primers from a different supplier.

Module 3:

1.Re-digestion, ligation and transformation

The plasmids were digested again, the target fragments were recovered, and the recovered fragments were re-ligated by Gibson assembly, then transformed into DH5α competent cells and plated, followed by cultivation at 37°C for 12 hours.





2.Colony PCR verification

Colony PCR system (10μL):

Green Mix	5μL
Primer-F	0.4μL
Primer-R	0.4μL
DDW	4.2μL
Colony	—

VanAB PCR program:

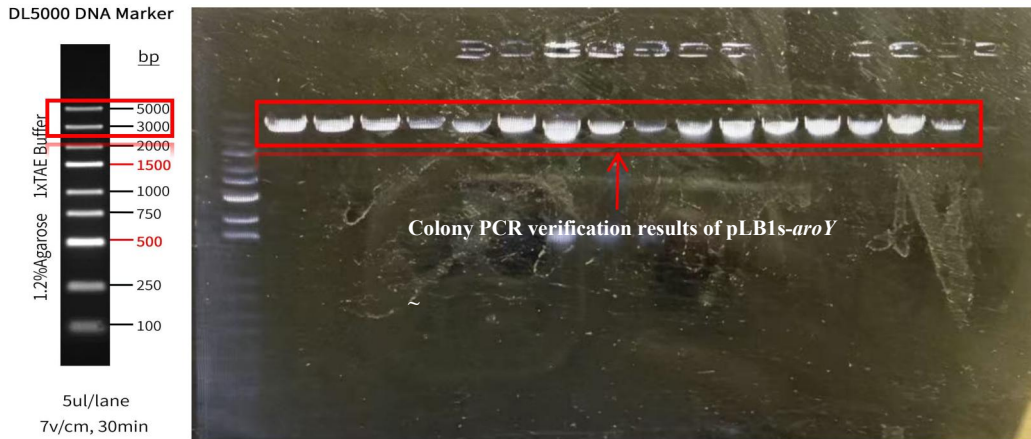
98°C	5min	} X30
98°C	30s	
58°C	30s	
72°C	65s	
72°C	5min	
25°C	∞	

aroY PCR program:

98°C	5min	
98°C	30s	} X11
-1.1°C	30s	
72°C	3m30s	
98°C	30s	} X20
60°C	30s	
72°C	3m30s	
25°C	∞	

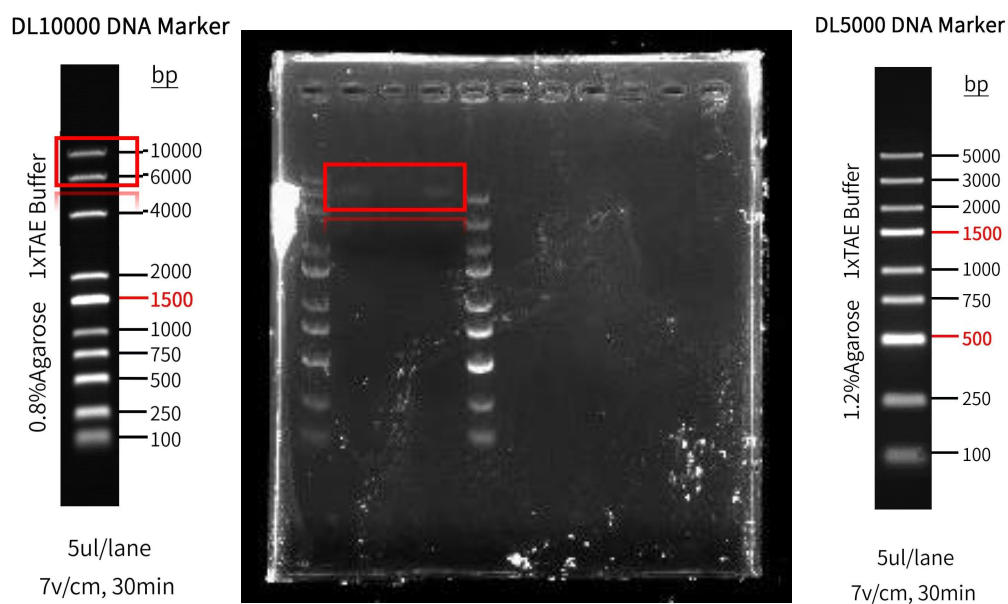
Single colonies were picked as templates for colony PCR, and the PCR products were verified by agarose gel electrophoresis.

The positive PCR products were excised and recovered from the gel.



3.Plasmid extraction, digestion and gel verification.

The verified clones were inoculated and shaken, then plasmids were extracted. The plasmids were digested and verified by agarose gel electrophoresis.



4. Induction:

Induction of BW-pLB1s-*VanAB-aroY*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
20% arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

5. Whole-cell catalysis

BW-pYB1a-*dmpB-dmpD-mhpD-mhpE* strain, with induced enzymes, catalyzes the conversion of VA to Pyr. BW-pLB1s-*VanAB-aroY* and BW-pYB1a-*dmpB-dmpD-mhpD-mhpE* strains express induced enzymes to catalyze the conversion of CA to Pyr. Determine the OD₆₀₀ of

induced bacterial cells using an ultraviolet spectrophotometer. Collect $OD_{600}=6$ units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in H_2SO_4 , and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at $30^{\circ}C$ for 1 hours.

The whole-cell catalytic system was configured as follows:

CA catalytic system (μL):

Strain	CA(20mM)	Fe2+(100mM)	Mn2+(100mM)	H2SO4(5mM)
BW-pYB1a-dmpB-dmpD-mh <i>pD-mhpE</i>	50	20	20	110
BW-pYB1a-dmpB-dmpD-mh <i>pD-mhpE</i>	50	0	0	150
BW-pYB1a-dmpB-dmpD-mh <i>pD-mhpE</i>	50	1	2	147
BW	50	20	20	110
BW	50	0	0	150
BW-pYB1a-eGFP	50	20	20	110
BW-pYB1a-eGFP	50	0	0	150

VA catalytic system (μL):

Strain	VA(10mM)	Fe2+(100mM)	Mn2+(100mM)	H2SO4(5mM)
BW-pLB1s-VanAB-aroY				
BW-pYB1a-dmpB-dmpD-mh <i>pD-mhpE</i>	100	0	0	100
BW-pLB1s-VanAB-aroY				
BW-pYB1a-dmpB-dmpD-mh <i>pD-mhpE</i>	100	1	2	97
BW	100	20	20	60
BW	100	0	0	100

Strain	VA(10mM)	Fe2+(100mM)	Mn2+(100mM)	H2SO4(5mM)
BW-pYB1a-eGFP	100	20	20	60
BW-pYB1a-eGFP	100	0	0	100

After catalysis, the system was centrifuged, filtered through 0.22 μ m membranes, and stored at -20°C.

6. High-Performance Liquid Chromatography (HPLC) Detection of Pyr

The mobile phase was prepared: 5mM H₂SO₄ aqueous solution; 5% acetonitrile/5mM H₂SO₄ and 30% acetonitrile/5mM H₂SO₄ were also prepared.

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

The samples were detected by HPLC using the sulfuric acid method (mobile phase: 5mM H₂SO₄) and the phosphoric acid method. Column: Aminex HPX-87H; flow rate: 0.6 mL/min; column temperature: 50°C; injection volume: 10 μ L.

We performed four HPLC detection on the samples, and the following are our test results.

